DEEPAKRAJ S

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SUMMARY

Software Engineer with 2 years of experience building backend systems using **Java, Spring Boot, microservices, REST APIs, Kafka, PostgreSQL, Redis, and AWS**. Experienced in developing scalable services, event-driven workflows, authentication systems, financial applications, database-intensive processing, and cloud deployments. Strong foundation in **data structures, object-oriented programming, multithreading, system design, distributed systems, database optimization, and backend architecture**.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, C, SQL
Backend: Spring Boot, Spring Framework, REST APIs, Microservices, Node.js
Distributed Systems: Apache Kafka, RabbitMQ, Event-Driven Architecture, Distributed Systems, Concurrency, Multithreading
Databases: PostgreSQL, MongoDB, Redis, SQL Optimization, Database Indexing
Cloud & DevOps: AWS, EC2, ECS, S3, Lambda, VPC, Docker, Terraform, GitLab CI/CD, Linux, Git
Observability: Prometheus, Grafana
AI/ML: Spring AI, Retrieval-Augmented Generation (RAG), Vector Databases, Qdrant, Pinecone, Agentic AI, Tool Calling, Model Context Protocol (MCP)
Core CS: Data Structures & Algorithms, Object-Oriented Programming, System Design, Operating Systems, Computer Networks

PROFESSIONAL EXPERIENCE

Software Engineer | Canopi India Private Limited *May 2026 – Present*
Bangalore, India

- Created lender compliance reports by aggregating data across 10+ modules, improving performance through database read streaming and enabling user-configurable cron scheduling with filters.
- Built DD APIs covering the complete transaction lifecycle, implementing validations, state transitions, and complex financial business rules and edge cases.
- Designed an end-to-end Spring Boot BFF to integrate with Liferay business APIs, implementing OAuth2, mTLS, payload encryption, authentication, and authorization.
- Developed a configurable **multi-step OTP authentication framework** supporting email and SMS verification for secure login workflows.
- Implemented Kafka-based push notifications with Firebase FCM, incorporating asynchronous processing, retry mechanisms, failure handling, and audit logging.

Software Engineer | Lightcast India Private Limited *January 2025 – May 2026*
Chennai, India

- Developed microservices handling **30K+ daily requests** across distributed production systems.
- Reduced API failure rates by **15%** through centralized request validation and structured exception handling.
- Improved service throughput by **20%** using Redis caching and PostgreSQL query optimization.
- Implemented **Kafka-based event streaming across 10+ services**, reducing inter-service communication latency by **30%**.
- Enhanced internal API authorization using **RBAC**, improving security compliance by **70%**.
- Provisioned AWS infrastructure using **Terraform**, improving deployment reliability by **25%**.
- Built **Prometheus and Grafana observability dashboards**, reducing incident response time by **40%**.
- Automated bulk data refresh workflows using **AWS EventBridge scheduled triggers**, reducing monthly developer maintenance effort by **70%**.

EDUCATION

Sona College of Technology, Salem *2021 – 2025*
Bachelor of Engineering in Computer Science and Engineering (CGPA: 8.9/10.0)

PROJECTS

Career Coach AI Agent — Intelligent Talent Guidance System

- Engineered a multi-tenant AI coaching platform using **springAI**, supporting **300+ active users** in a production environment.
- Implemented **tool calling, context management, retrieval workflows, and latency optimization** to improve agent responsiveness and platform efficiency.

E-Commerce Distributed Transaction Platform

- Engineered a Spring Boot microservices architecture supporting 10K+ daily transactions, using idempotency keys and deduplication checks to guarantee exactly-once payment processing across services.
- Designed a retry and reconciliation pipeline with exponential backoff for failed Kafka events, reconciling order and payment state to maintain 99.9% data consistency across distributed services.
- Integrated **Eureka Service Discovery, API Gateway, JWT authentication,** PostgreSQL, Docker, and Kafka across 5+ services, reducing service configuration overhead by 40%.

CERTIFICATIONS

AWS Solutions Architect – Associate • AWS Cloud Practitioner • GitLab CI/CD Certification • Docker Certification • Agentic AI • From Java Developer to AI Engineer • Software Engineering • Data Structures in Java • Python Programming • C Programming • Network Essentials